Project Proposal: Dreaming as Synthetic Replay

COMS 5730: Machine Learning — Group 2

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Motivation

During sleep, our brains provide distorted, recombined replay of waking experiences in the form of dreams. Some theories propose that this process contributes to memory consolidation and generalization. Along these lines, Erik Hoel's "Overfitted Brain Hypothesis" suggests that dreaming provides altered versions of frequent experiences in order to prevent overfitting.

An analogous idea can be tested with machine learning since, as a data model, neural networks also suffer from overfitting. Many existing generalization techniques work by preventing the model from fitting individual training samples too precisely. We will investigate whether a separate "dreaming" phase of training on distorted and recombined data can improve generalization.

The primary question is: what properties must synthetic replay have in order to improve neural network generalization? The main hypothesis is that as distortion increases, and the original information that the data represents is lost, a model prone to overfitting will improve in its predictive capability before deteriorating due to excessive, meaningless distortion. This suggests the possibility of an optimum that produces the strongest generalization.

Methodology

An image classification problem seems to be the most fitting test environment. MNIST datasets and potentially CIFAR-10 will be used.

A network will first undergo conventional training with a set size and model capacity which will produce measurable overfitting. After this, the model will undergo a dream phase during which various replay strategies will be employed. These might include noise, transformations, or more complex recombinations like interpolations.

The primary experiments will seek to quantify the effect of dream distortion strength on model performance for the varying distortion methods. We may also examine whether the value of synthetic replay increases for limited-data regimes.